

WEEKLY ASSIGNMENT-1

-Submitted by Anusha Kucharlapati

CLIENT MASTER			PRODUCT MASTER			SALESMAN MASTER		
CLIENTNO	vc2(6)	PK & Must start with 'C'	PRODUCTNO	vc2(6)	PK & Must start with 'P'	SALESMANNO	vc2(6)	PK & Must start with 'S'
NAME	vc2(20)	NOT NULL	DESCRIPTION	vc2(15)	NOT NULL	SALESMANNAME	vc2(20)	NOT NULL
ADDRESS1	vc2(30)		PROFITPERC	n(4,2)	NOT NULL	ADDRESS1	vc2(30)	NOT NULL
ADDRESS2	vc2(30)		UNITMEASURE	vc2(10)	NOT NULL	ADDRESS2	vc2(30)	
CITY	vc2(15)		QTYONHAND	n(8)	NOT NULL	CITY	vc2(20)	
PINCODE	n(8)		REORDERLVL	n(8)	NOT NULL	PINCODE	n(8)	
STATE	vc2(15)		SELLPRICE	n(8,2)	NOT NULL & can't be 0	State	vc2(20)	
BALDUE	n(10,2)		COSTPRICE	n(8,2)	NOT NULL & can't be 0	SALAMT	n(8,2)	NOT NULL & Can't be 0
						TGTOGET	n(6,2)	NOT NULL
						YTDSALES	n(6,2)	NOT NULL
						REMARKS	vc2(60)	
SALES ORDER			SALES ORDER DETAILS					
ORDERNO	vc2(6)	PK & Must start with 'O'	ORDERNO	vc2(6)	FK to SALES_ORDER			
CLIENTNO	vc2(6)	FK to CLIENT_MASTER	PRODUCTNO	vc2(6)	FK to PRODUCT_MASTER			
ORDERDATE	date		QTYORDERED	n(8)				
DELYADDR	vc2(25)		QTYDISP	n(8)				
SALESMANNO	vc2(6)	FK to SALESMAN_MASTER	PRODUCTRATE	n(10,2)				
DELYTYPE	char(1)	Check 'P' or 'F'	Create Composite PRIMARY KEY of ORDERNO and PRODUCTNO					
BILLEDYN	char(1)	Check 'Y' or 'N'						
DELYDATE	date							
ORDERSTATUS	vc2(10)	Check 'In Process' or 'Fulfilled' or 'Backorder' or 'Cancelled'						

Fig 1 Tables Description

1. Create Table Commands

--Client_Master

```
CREATE TABLE Client_Master (
    ClientNo VARCHAR(6) CHECK (ClientNo LIKE 'C%'),
    Name VARCHAR(20) NOT NULL,
    Address1 VARCHAR(30),
    Address2 VARCHAR(30),
    City VARCHAR(12),
    Pincode DECIMAL(8,0),
    State VARCHAR(15),
    BalDue DECIMAL(10,2),
    PRIMARY KEY (ClientNo)
);
```

--Product_Master

```
CREATE TABLE Product_Master (
    ProductNo VARCHAR(6) CHECK (ProductNo LIKE 'P%'),
    Description VARCHAR(50) NOT NULL,
    ProfitPercent DECIMAL(4,2) NOT NULL,
    UnitMeasure VARCHAR(10) NOT NULL,
    QtyOnHand DECIMAL(8,0) NOT NULL,
    ReorderLvl DECIMAL(8,0) NOT NULL,
    SellPrice DECIMAL(8,2) NOT NULL CHECK (SellPrice > 0),
    CostPrice DECIMAL(8,2) NOT NULL CHECK (CostPrice > 0),
    PRIMARY KEY (ProductNo)
);
```

--Salesman_Master

```
CREATE TABLE Salesman_Master (
    SalesmanNo VARCHAR(6)
        CHECK (SalesmanNo LIKE 'S%'),
    SalesmanName VARCHAR(20) NOT NULL,
    Address1 VARCHAR(30) NOT NULL,
```

```

Address2 VARCHAR(30),
City VARCHAR(20),
Pincode DECIMAL(8,0),
State VARCHAR(20),
SalAmt DECIMAL(8,2) NOT NULL
    CHECK (SalAmt <> 0),
TgtToGet DECIMAL(6,2) NOT NULL,
YtdSales DECIMAL(6,2) NOT NULL,
Remarks VARCHAR(60),
PRIMARY KEY (SalesmanNo)
);

```

--Sales_Order

```

CREATE TABLE Sales_Order (
    OrderNo VARCHAR(6) CHECK (OrderNo LIKE '0%'),
    ClientNo VARCHAR(6) NOT NULL,
    OrderDate DATE,
    DelyAddr VARCHAR(25),
    SalesmanNo VARCHAR(6) NOT NULL,
    DelyType CHAR(1) CHECK (DelyType IN ('P', 'F')),
    BilledYN CHAR(1) CHECK (BilledYN IN ('Y', 'N')),
    DelyDate DATE,
    OrderStatus VARCHAR(10)
    CHECK (OrderStatus IN ('In
Process', 'Fulfilled', 'Backorder', 'Cancelled')),
    PRIMARY KEY (OrderNo),
    FOREIGN KEY (ClientNo) REFERENCES Client_Master(ClientNo),
    FOREIGN KEY (SalesmanNo) REFERENCES Salesman_Master(SalesmanNo)
);

```

--Sales_Order_Details

```

CREATE TABLE Sales_Order_Details (
    OrderNo VARCHAR(6),
    ProductNo VARCHAR(6),
    QtyOrdered DECIMAL(8,0),
    QtyDisp DECIMAL(8,0),
    ProductRate DECIMAL(10,2),
    PRIMARY KEY (OrderNo, ProductNo),
    FOREIGN KEY (OrderNo) REFERENCES Sales_Order(OrderNo),
    FOREIGN KEY (ProductNo) REFERENCES Product_Master(ProductNo)
);

```

2. Insert Commands

```

INSERT INTO Client_Master
(ClientNo, Name, Address1, Address2, City, Pincode, State, BalDue)
VALUES
('C00001', 'Ivan Bayross', 'MG Road', 'Bombay', 'Mumbai', 400054, 'Maharashtra',
15000.00),
('C00002', 'Mamta Muzumdar', 'Madras', '', 'Chennai', 780001, 'Tamil Nadu',
0.00),
('C00003', 'Chhaya Bankar', 'Bombay', '', 'Mumbai', 400057, 'Maharashtra',
5000.00),
('C00004', 'Ashwini Joshi', 'Bangalore', '', 'Bangalore', 560001, 'Karnataka',
0.00);

```

```
INSERT INTO Salesman_Master VALUES
('S00001', 'Aman', 'A/14', 'Worli', 'Mumbai', 400002, 'Maharashtra', 3000.00, 100.00,
50.00, 'Good'),
('S00002', 'Omkar', '65', 'Nariman', 'Mumbai', 400001, 'Maharashtra', 3000.00, 200.00,
100.00, 'Good'),
('S00003', 'Raj', 'P-
7', 'Bandar', 'Mumbai', 400032, 'Maharashtra', 3000.00, 300.00, 200.00, 'Good'),
('S00004', 'Ashish', 'A/5', 'Juhu', 'Mumbai', 400044, 'Maharashtra', 3500.00, 200.00,
150.00, 'Good');
```

```
INSERT INTO Product_Master VALUES
('P00001', 'T-Shirt', 5.00, 'Piece', 200, 50, 350.00, 250.00),
('P00002', 'Shirt', 6.00, 'Piece', 150, 40, 500.00, 350.00),
('P00003', 'Jeans', 5.50, 'Piece', 100, 30, 800.00, 600.00),
('P00004', 'Jacket', 7.00, 'Piece', 80, 20, 1200.00, 900.00);
```

```
INSERT INTO Sales_Order VALUES
('019001', 'C00001', '2023-06-12', 'Mumbai', 'S00001', 'P', 'Y', '2023-06-
15', 'Fulfilled'),
('019002', 'C00002', '2023-06-13', 'Chennai', 'S00002', 'F', 'N', '2023-06-18', 'In
Process'),
('019003', 'C00003', '2023-06-14', 'Mumbai', 'S00003', 'P', 'Y', '2023-06-
20', 'Fulfilled'),
('019004', 'C00004', '2023-06-15', 'Bangalore', 'S00004', 'F', 'N', '2023-06-
22', 'Backorder');
```

```
INSERT INTO Sales_Order_Details VALUES
('019001', 'P00001', 10, 10, 350.00),
('019002', 'P00002', 5, 3, 500.00),
('019003', 'P00003', 7, 7, 800.00),
('019004', 'P00004', 4, 2, 1200.00);
```

3.1 Display names of all Clients.

```
SELECT Name from Client_master;
```

Results		Messages	
	Name		
1	Ivan Bayross		
2	Mamta Muzumdar		
3	Chhaya Bankar		
4	Ashwini Joshi		
5	Ravi Kumar		
6	Neha Verma		
7	Old Client		

3.2 Display all Clients who are Located in Mumbai

```
SELECT *  
FROM Client_Master  
WHERE City = 'Mumbai';
```

	ClientNo	Name	Address1	Address2	City	State	BalDue	Phone_no	Pincode
1	C00001	Ivan Bayross	MG Road	Bombay	Mumbai	Maharashtra	15000.00	NULL	400054
2	C00003	Chhaya Bankar	Bombay		Mumbai	Maharashtra	5000.00	NULL	400057

3.3 Display all products whose selling price is <5000 and >2000.

```
SELECT *  
FROM Product_Master  
WHERE SellPrice > 2000 AND SellPrice < 5000
```

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLvl	SellPrice	CostPrice
1	P00006	Printer	6.00	Piece	50	10	4500.00	3800.00
2	P00007	Trousers	6.50	Piece	120	30	2200.00	1600.00

3.4 Display Name, City, State of Clients not in the state of Maharashtra

```
SELECT Name, City, State  
FROM Client_Master  
WHERE State <> 'Maharashtra';
```

	Name	City	State
1	Mamta Muzumdar	Chennai	Tamil Nadu
2	Ashwini Joshi	Bangalore	Karnataka
3	Ravi Kumar	Hyderabad	Telangana
4	Old Client	Delhi	Delhi

3.5 Display all the information of ClientNo C0001 and C0002.

```
SELECT *  
FROM Client_Master  
WHERE ClientNo IN ('C00001', 'C00002');
```

	ClientNo	Name	Address1	Address2	City	State	BalDue	Phone_no	Pincode
1	C00001	Ivan Bayross	MG Road	Bombay	Mumbai	Maharashtra	15000.00	NULL	400054
2	C00002	Mamta Muzumdar	Madras		Chennai	Tamil Nadu	0.00	NULL	780001

3.6 Change the selling price of '1.44 drive' to 1150.50

```
UPDATE Product_Master  
SET SellPrice = 1150.50  
WHERE Description = '1.44 drive';
```

	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLvl	SellPrice	CostPrice
1	P00005	1.44 drive	5.00	Piece	100	20	1150.50	800.00

3.7 Delete the record of ClientNo C0005

```
DELETE FROM Client_Master  
WHERE ClientNo = 'C00005';
```

3.8 Display the clients who stay in a city whose second letter is 'a'.

```
SELECT *  
FROM Client_Master  
WHERE City LIKE '_a%';
```

Results Messages									
	ClientNo	Name	Address1	Address2	City	State	BalDue	Phone_no	Pincode
1	C00004	Ashwini Joshi	Bangalore		Bangalore	Karnataka	0.00	NULL	560001

3.9 Count the number of products having selling price greater than or equal to 1500.

```
SELECT COUNT(*) AS ProductCount  
FROM Product_Master  
WHERE SellPrice >= 1500;
```

Results Messages	
	ProductCount
1	3

3.10 Display the order number, product number, quantity ordered, quantity dispatched, and balance quantity (balance quantity is not stored in the table).

```
SELECT  
    QtyOrdered,  
    QtyDisp,  
    (QtyOrdered - QtyDisp) AS BalancedQty  
FROM Sales_Order_Details;
```

	QtyOrdered	QtyDisp	BalancedQty
1	10	10	0
2	5	3	2
3	7	7	0
4	4	2	2
5	3	1	2
6	8	0	8
7	8	0	8
8	3	0	3

4. Queries on JOINS

4.1 Find the products sold to the client "Ivan Bayross".

```
select p.ProductNo, p.Description from Client_Master c  
join Sales_Order s on s.ClientNo=c.ClientNo  
join Sales_Order_Details d on d.OrderNo=s.OrderNo  
join Product_Master p on d.ProductNo=p.ProductNo  
where c.Name='Ivan Bayross';
```

Results Messages		
	ProductNo	Description
1	P00001	T-Shirt
2	P00006	Printer
3	P00007	Trousers

4.2 Find the products and their quantities that are to be delivered in the current month.

```
select p.ProductNo, p.Description, s.DelyDate, s.OrderStatus from Sales_Order s
join Sales_Order_Details d on d.OrderNo=s.OrderNo
join Product_Master p on d.ProductNo=p.ProductNo
where month(s.DelyDate)=1 and s.OrderStatus in ('In Process', 'Backorder');
```

	ProductNo	Description	DelyDate	OrderStatus
1	P00005	1.44 drive	2026-01-15	In Process
2	P00006	Printer	2026-01-18	In Process
3	P00007	Trousers	2026-01-18	In Process
4	P00008	Pull Overs	2026-01-20	In Process

4.3 Find the products that are constantly sold (rapidly moving products).

```
select p.ProductNo, p.Description, sum(d.QtyOrdered) as Total_qty_sold from
Sales_Order_Details d
join Product_Master p on d.ProductNo=p.ProductNo
group by p.ProductNo, p.Description
having sum(d.QtyOrdered)=(Select top 1 sum(d.QtyOrdered) from Sales_Order_Details
d
group by d.ProductNo
order by sum(d.QtyOrdered) desc);
```

	ProductNo	Description	Total_qty_sold
1	P00001	T-Shirt	10

4.4 Find the names of clients who have purchased the product “Trousers”.

```
select c.Name from Client_Master c
join Sales_Order s on s.ClientNo=c.ClientNo
join Sales_Order_Details d on d.OrderNo=s.OrderNo
join Product_Master p on d.ProductNo=p.ProductNo
where p.Description='Trousers';
```

	Name
1	Ivan Bayross

4.5 Find the products and orders where less than 5 units of the product “Pull Overs” are ordered.

```
select s.ClientNo, p.ProductNo, p.Description, d.OrderNo, d.QtyOrdered from
Sales_Order s
join Sales_Order_Details d on d.OrderNo=s.OrderNo
join Product_Master p on d.ProductNo=p.ProductNo
where p.Description='Pull Overs' and d.QtyOrdered<5;
```

	ClientNo	ProductNo	Description	OrderNo	QtyOrdered
1	C00006	P00008	Pull Overs	O19007	3

5. SUBQUERIES

5.1 Find the products that have not been sold so far.

```
select * from Product_Master  
where ProductNo not in (select ProductNo from Sales_Order_Details);
```

Results		Messages						
	ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLvl	SellPrice	CostPrice
1	P00009	Cap	5.00	Piece	150	40	500.00	300.00

5.2 Find the name and address of the client who placed order number O19001.

```
select c.Name, c.Address1+', '+c.Address2 as Address from Client_Master c  
where c.ClientNo=(select ClientNo from Sales_Order where OrderNo='O19001');
```

	Name	Address
1	Ivan Bayross	MG Road, Bombay

5.3 Find the clients who have placed orders before May 2002.

```
select c.ClientNo, c.Name from Client_Master c  
where ClientNo in (select ClientNo from Sales_Order s  
where s.OrderDate < '2025-05-02');
```

	ClientNo	Name
1	C00001	Ivan Bayross
2	C00002	Mamta Muzumdar
3	C00003	Chhaya Bankar
4	C00004	Ashwini Joshi
5	C00007	Old Client

6. Write commands for the following

6.1 Display the system date in the format: Saturday, February 11, 2012

```
select cast(sysdatetime() as date);
```

	(No column name)
1	2026-01-03

6.2 Display the balance due from Client_Master table in the format \$99,999.99.

```
SELECT ClientNo,  
        FORMAT(BalDue, 'C') AS BalanceDue  
FROM Client_Master;
```

	ClientNo	BalanceDue
1	C00001	\$15,000.00
2	C00002	\$0.00
3	C00003	\$5,000.00
4	C00004	\$0.00
5	C00005	\$2,000.00
6	C00006	\$0.00
7	C00007	\$0.00

6.3 Display the following message: “Salesman Aman sold goods of 50 while given target was 100”

```
SELECT  
'Salesman ' + SalesmanName +  
' sold goods of ' + CAST(YtdSales AS VARCHAR(10)) +  
' while given target was ' + CAST(TgtToGet AS VARCHAR(10)) AS Message  
FROM Salesman_Master  
WHERE SalesmanName = 'Aman';
```

	Message
1	Salesman Aman sold goods of 50.00 while given tar...

6.4 Display your age in years.

```
select datediff(year, '2005-07-02', getdate()) as age;
```

	age
1	21